BrahmaPath: An Offline-First Indigenous Psychometric State Machine and Digital Therapeutics Platform for Civilian Focus and Frontline Defense Equilibrium

Standardized Implementation Repository: https://github.com/aayushm-stack/IKO-103_Project

Abstract

Modern digital self-regulation technologies suffer from a persistent paradox: commercial habit apps enforce rigid, punitive streak counters that induce the Abstinence Violation Effect, while existing cultural and spiritual applications remain passive textual dictionaries or unconstrained, cloud-reliant conversational wrappers. This paper introduces **BrahmaPath**, an offline-first, deterministic psychometric state machine and digital therapeutics platform engineered to bridge classical Indian Knowledge Systems (IKS) with contemporary computational behavior design. Grounded in the *Ratha-Kalpana* (Chariot Allegory) of the *Katha Upanishad*, the tripartite *Trayopastambha* physiology of the *Charaka Samhita*, the eight-stage impulse cascade of *Bhagavad Gita* 2.62–63, and the cognitive restructuring frameworks of *Patanjali's Yoga Sutras* (1.33 and 2.33), **BrahmaPath** operationalizes scriptural psychology into continuous, real-time mathematical feedback. The platform implements a dual-stage, six-cohort micro-curation architecture serving civilian daily rhythms (Learning & Study, Work & Enterprise, Home & Quiet Living) and frontline defense posting terrains (High-Altitude/Extreme Cold, Active Border/Picket, Cantonment/Peace Station). We specify the closed-form mathematical equations governing the five interdependent engines: the 100-point continuous Trayopastambha Yukta Score (Y), the Pratipaksha Resilience Quotient (P), the Bhāva Emotional Buffer Score (B), the Master Chariot Composure Index (C), and the non-punitive -5% natural Tamasic drift momentum model (M). We document a chronological full-day behavioral simulation demonstrating real-time reactivity to slips and cognitive reframing, present a cross-stage schema isolation proof (16 vs. 22 inputs), detail the 7-day rolling Android OS alarm queue resolving native repetition failures, and articulate our foundational architectural decision to reject generative AI chatbots in favor of deterministic clinical safety and 100% offline military operational security (OPSEC).

Keywords: Digital Therapeutics, Indian Knowledge Systems (IKS), Psychometric State Machine, Dual-Systems Theory, Cognitive Friction, Military Psychiatry, Offline-First Architecture.

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1 Introduction, Motivation, and Target Audience

1.1 The Universal Human Need

The challenge of directing attention, moderating biological rhythms, and preserving emotional equilibrium in the presence of recurring internal agitation is universal across human vocations. Whether an undergraduate student struggles against compulsive algorithmic social media feeds during late-night exam revision, a software engineer manages cognitive exhaustion across back-to-back corporate sprints, a retired elder seeks peaceful afternoon rhythm, or an infantry sentinel endures sub-zero vigilance at 18,000 feet on the Siachen glacier, the underlying neuro-cognitive architecture is identical. The senses react impulsively to immediate sensory objects (*Preyas*), hijacking conscious cognitive discernment (*Buddhi*) and fracturing inner composure. **BrahmaPath** is engineered for **any person seeking greater inner peace, a calm personal life outside of work, or sustained focus and vitality in daily duty** by translating profound, centuries-tested Indian philosophical frameworks into structured daily behavioral habits.

1.2 The Dual Pathology of Existing Solutions

Contemporary software solutions intended to support mental focus and stress regulation divide into two failed paradigms:

1. **Punitive Western Habit Trackers and Streak Blockers:** Modern focus applications (e.g., rigid timers, website blockers, and binary day-counter streak apps) rely on coercive, extrinsic suppression. As demonstrated by human-computer interaction (HCI) literature [1], rigid barriers fail immediately once deactivated because they target automatic impulses without cultivating intrinsic discriminative mastery. Crucially, when an inevitable disruption occurs—such as a military night deployment, family emergency, or brief digital distraction—binary streak systems drop the counter to zero. This triggers deep psychological guilt, distress (*Viṣāda*), and the clinically recognized *Abstinence Violation Effect* [1], causing over 70% of users to abandon the intervention entirely.
2. **Passive Scripture Dictionaries and Chatbot Wrappers:** Existing scriptural wellness platforms (e.g., *Situational Gita*, *Living Geeta*, *Ask Krishna Ji*) function as static verse repositories or superficial prompt wrappers around commercial Large Language Models (LLMs). When a user experiences acute stress or digital cravings, reading an abstract verse or engaging in an ungrounded chat produces fleeting intellectual agreement without altering somatic state, circadian rest, or physical impulse pathways. Furthermore, cloud-dependent LLM chatbots introduce unacceptable risks of theological hallucination, lack clinical crisis safeguards, and violate military Operational Security (OPSEC) by streaming sensitive personnel data to remote corporate servers.

1.3 The Kurukshetra Premise: Two Stages and Six Contextual Cohorts

The foundational thesis of **BrahmaPath** is anchored in the historical setting of the *Bhagavad Gita*: the profoundest psychological manual on inner equilibrium was not delivered in an isolated forest hermitage, but directly upon the high-stakes battlefield of Kurukshetra. Consequently, during initial onboarding (Page 2), the system explicitly presents two operational paths of action (*Svadharma*), branching into six micro-curated human environments:

- **Stage 1: The Inner Battleground (Path of the Seeker — Civilian Rhythms):**
    1. *Learning & Study Cohort:* Designed for university students, researchers, and competitive exam aspirants navigating algorithmic doomscrolling, peer placement envy, and academic overwhelm.
    2. *Work & Enterprise Cohort:* Designed for corporate professionals, executives, and entrepreneurs navigating decision fatigue, executive burnout, high stimulant/caffeine intake, and evening boundary decay.
    3. *Home & Quiet Living Cohort:* Designed with complete dignity and zero slashes for homemakers, caregivers, and retired seniors managing domestic anxiety, gentle digestive health, and quiet afternoon rhythm.
- **Stage 2: The Frontline Battleground (Path of the Sentinel — Defense Terrains):**
    4. *High-Altitude / Extreme Cold Terrain:* Tailored for soldiers deployed in extreme sub-zero fiberglass bunkers (Siachen, Eastern Ladakh, 15,000–20,000+ ft), addressing hypoxia-induced loss of appetite (*Agni-Māndya*), disturbed hypoxic sleep, and family helplessness.

5. *Active Border / Picket Terrain:* Tailored for infantry and Border Security Force (BSF) personnel stationed along active fences (e.g., LOC, international borders), mitigating six-cup sweet watch-tea caffeine overdrive, broken split-sentry sleep, and tense phone calls from home.
6. *Cantonment / Peace Station Terrain:* Tailored for personnel in peace stations, regimental centers, and headquarters, addressing inspection stress, swallowed hierarchy friction, and leave-delay guilt.

1.4 Database Integrity and Schema Isolation Rule

Civilian cognitive habits and military operational conditions require fundamentally distinct data structures. Stage 1 captures 16 discrete user interface inputs mapped to 14 database schema fields, whereas Stage 2 captures 22 interface inputs mapped to 19 specialized telemetry fields (incorporating sentry ration profiles, tactical watch-sleep protocols, and peer battle-buddy cohesion). Because mixing these schemas within an active trajectory corrupts continuous psychometric state calculations, **BrahmaPath** enforces strict schema isolation: users cannot casually toggle between Civilian and Defense paths in user settings. Transitioning between stages requires an explicit Factory Reset in the Settings "Danger Zone", wiping local key-value stores and initializing an unpolluted baseline.

2 Literature Review, Prior Art, and Systematic Derivation

To establish rigorous scientific grounding, the architecture of **BrahmaPath** was systematically derived from four research domains: human-computer interaction, military psychiatry, clinical psychometrics, and digital health patent jurisprudence.

2.1 Dual-Systems Theory and Cognitive Friction in HCI

In their comprehensive survey of 367 digital self-control tools, Lyngs et al. [1] applied Kahneman's Dual-Systems Theory to demonstrate that external blockers only suppress automatic System 1 processes. Once the timer expires, the underlying compulsive loop reactivates with increased intensity. To achieve enduring self-regulation, digital interventions must introduce *design friction* that slows impulsive execution and engages conscious, reflective System 2 cognitive deliberation. This empirical finding directly shaped our **Impulse Interceptor Engine**, which replaces binary lockouts with a seven-stage cognitive deceleration gateway and an active *Vitarka-Pratipaksha* reframing protocol.

2.2 High-Altitude Military Physiology and Homefront Distress

Deployments in extreme high-altitude areas (EHAA, 15,000–20,000+ ft) introduce severe physiological and psychological stressors. In a definitive clinical study of 334 soldiers deployed in EHAA, Rajesh, Bhat, Srivastava, and Das [2] proved that environmental hypoxia induces high rates of anorexia, somatic nausea, and severely fragmented sleep architecture. Statistically, married service personnel exhibited significantly higher psychological morbidity on the General Health Questionnaire (*GHQ-12*) compared to unmarried peers ($p < 0.01$). This vulnerability is driven not by battlefield fear, but by acute helplessness over domestic disputes and family crises occurring thousands of kilometers away while isolated in sub-zero fiberglass huts. This clinical evidence directly generated our **High-Altitude Terrain Engine**, the **Jaṭharāgni Ration & Hydration Check** in 'SentinelSadhana.tsx' (which triggers warm-water and soup

micro-prompts when nausea suppresses appetite), and the **Homefront Reassurance Bridge** in 'SentinelCompass.tsx'.

2.3 The "Last Straw" Phenomenon and Border Fatigue

Investigating non-combat operational attrition across the Armed Forces, Srivastava et al. [3] and longitudinal BSF paramilitary studies by Chhabra & Chhabra [12] demonstrated that fratricidal surges and stress breakdowns rarely originate from isolated catastrophic incidents. Rather, months of cumulative circadian sleep debt (36.3%), irregular mess timings and heavy sentry-tea dependence (36.0%), and suppressed hierarchy resentment progressively deplete the soldier's inner coping capacity. When this reserve is exhausted, an innocuous remark from a peer or a minor domestic argument acts as the catastrophic *"last straw."* In response, **BrahmaPath** engineered the **Inner Coping & Patience Reserve (*Ojas* Tank 1–5 slider)**, the **Watch-Tea Stimulant Profile Check**, and the **10-Minute *Vāṅ*-maya Tapas Speech Shield** (BG 17.15) to break the escalation loop before speech occurs.

2.4 Military Family Resilience and Peer Cohesion

Studies by Chatterjee et al. [4], Pawar et al. [6], and the Armed Forces Medical Services Concept on Holistic Fitness [5] confirm that peer camaraderie (*Sahyog*) and structured family communication are the primary psychological buffers protecting armed service personnel. When institutional mental health tools offer cold, individualistic advice (e.g., "focus only on your mission and detach from home"), personnel reject them as unfeeling. **BrahmaPath** replaces clinical detachment with warm **Sahyogi (Battle-Buddy) Psychology**, integrating a pre-scripted 2-Sentence Homefront Reassurance script, pre-call de-armoring exercises, and peer debrief telemetry.

2.5 Clinical IKS Validation and Tri-Guna Psychometrics

The application of ancient Indian psychological concepts to contemporary cognitive psychotherapy is clinically established. M. S. Reddy [7] systematically mapped the dialogue between Krishna and Arjuna in the *Bhagavad Gita* to modern crisis Cognitive Behavioral Therapy (CBT), identifying verses 6.26, 2.14, 12.13, 2.56, and 6.5 as precise cognitive reframing interventions for acute anxiety, grief, and emotional paralysis. Simultaneously, Shilpa and Murthy [8] developed and standardized the *Mysore Triguna Scale*, confirming that the classical personality dimensions of *Sattva* (clarity, harmony), *Rajas* (restless agitation, passion), and *Tamas* (inertia, confusion) exhibit high psychometric reliability and construct validity. **BrahmaPath** translates these static pencil-and-paper psychometrics into a real-time, deterministic software state machine ('gunaDoshaEngine.ts').

2.6 Patent Prior Art and TKDL Non-Preemption Analysis

A formal patent clearance search was conducted across USPTO, Derwent Innovation, WIPO, and Indian Patent Office (InPASS) databases under Cooperative Patent Classifications (CPC) **G16H 20/70** (ICT for psychological therapy), **G16H 20/90** (ICT for traditional medicine), and **G16H 50/30** (ICT for health risk assessment). Table 1 delineates **BrahmaPath**'s structural non-infringement and technical novelty against granted prior art.

Traditional Knowledge Digital Library (CSIR-TKDL) Alignment: The Traditional Knowledge Digital Library (TKDL), established jointly by CSIR and the Ministry of AYUSH, protects traditional Indian formulations (such as medicinal plant extracts) and physical Hatha Yoga

Table 1: Patent Landscape and Prior Art Comparison

Document / App	Assignee / Origin	Disclosed Architecture	BrahmaPath Inventive Distinction
US 2024/0008813 A1 (IN 202141012087 A)	AyurAi Pvt. Ltd. (B. Pesala, 2024)	Hardware pulse/photoplethysmography sensor for Ayurvedic Dosha estimation.	Purely software-driven behavioral telemetry; zero external hardware; real-time relative screen and sentry-rest inputs.
US 11,617,746 B1 (Granted Patent)	Pear Therapeutics, Inc. (Granted 2023)	Cloud-dependent digital CBT platform for clinical depression.	100% offline OPSEC architecture for defense forward posts; indigenous Trayopastambha (34+33+33) and Gita 2.62 state machine.
US 2021/0098110 A1 (Published App)	P. Periyasamy et al. (Published 2021)	Absolute screen-time threshold monitoring triggering audio relaxation.	Self-Anchored 4-Zone Relative Screen Baseline ($r = M_{\text{digital}}/M_{\text{baseline}}$); rewards downward momentum without binary lockouts.
Situational Gita **Living Geeta** (Apps)	Antarjaal (2017) Living Geeta (2024)	Static categorised verse dictionaries for general life challenges.	Passive text retrieval only; zero behavioral habit tracking, biological restoration scoring, or impulse circuit-breakers.
Ask Krishna Ji (Consumer LLM App)	Commercial Generative AI App (2023)	Conversational prompt wrapper querying commercial cloud LLM APIs.	Rejects generative chatbots; zero hallucination; 100% deterministic scriptural validation; zero cloud data transmission.

postures from biopiracy. TKDL does not preempt deterministic software engineering algorithms, multi-engine mathematical weighting, or digital state machines operating under CPC G16H 20/70. Rather, **BrahmaPath**'s formal translation of traditional psychology directly operationalizes the national mandate of the **Indian Knowledge System & Mental Health Applications (IKSMHA) Centre at IIT Mandi**.

3 Core Mathematical Models and Psychometric State Machine

The computational core of **BrahmaPath** consists of five interconnected mathematical engines, operating entirely on-device without cloud network round-trips.

3.1 Engine 1: Trayopastambha Yukta Score Engine

Rooted in *Charaka Samhita* (Sutrasthana 11.35) and *Bhagavad Gita* 6.17, the Yukta Engine evaluates physiological and attentional equilibrium ($Y \in [0, 100]$) across the three biological sub-pillars: *Āhāra* (nourishment and sensory intake), *Nidrā* (circadian sleep recovery), and *Brahmacharya* (vital energy conservation and deep focus):

$$Y = S_{\bar{\text{A}}\text{hāra}} + S_{\text{Nidrā}} + S_{\text{Brahmacharya}} \tag{1}$$

where $S_{\bar{\text{A}}\text{hāra}} \in [0, 34]$, $S_{\text{Nidrā}} \in [0, 33]$, and $S_{\text{Brahmacharya}} \in [0, 33]$. If an individual pillar has not yet been logged today by the user, its score defaults strictly to 0 via the `loggedPillars` tracking interface.

3.1.1 Āhāra Pillar Formulation (Max 34 Points)

$$S_{\bar{\text{A}}\text{hāra}} = \min(34,\ P_{\text{nutrition}} + P_{\text{screen}} + P_{\text{mindful}}) \tag{2}$$

- **Nutrition / Ration Score ($P_{\text{nutrition}} \in \{2, 6, 10\}$):** For Stage 1 civilian users, fresh Sattvic intake awards 10, Rajasic stimulant meals award 6, and stale/junk Tamasic food awards 2. For Stage 2 defense sentries, a balanced operational ration awards 10, high watch-caffeine tea awards 6, and irregular/missed rations award 2.
- **Self-Anchored Relative Screen Time ($P_{\text{screen}} \in [0, 12]$):** Rather than applying arbitrary population cutoffs (e.g., "2 hours per day"), the engine anchors against the user's personal baseline $M_{\text{baseline}} = \max(60, \texttt{baselineMinutesAtLog})$, computing the relative ratio $r = \frac{M_{\text{digital}}}{M_{\text{baseline}}}$. The score is awarded via a smooth piecewise-linear curve:

$$P_{\text{screen}}(r) = \begin{cases} \text{round}(12 - 2.67r), & r \le 0.75 \quad \text{(Zone 1: Significant Drop, } 12 \to 10\text{)} \\ \text{round}(10 - 12(r - 0.75)), & 0.75 < r \le 1.00 \quad \text{(Dropped/Maintained, } 10 \to 7\text{)} \\ \max(0, \text{round}(7 - 11.67(r - 1.00))), & r > 1.00 \quad \text{(Zone 4: Above Average Taper, } 7 \to 0\text{)} \end{cases} \tag{3}$$

- **Sensory Mindfulness / Call Digestion ($P_{\text{mindful}} \in [0, 12]$):** Computed continuously from the 1–5 interface slider x:

$$P_{\text{mindful}}(x) = \max(0, \min(12, \text{round}((x - 1) \times 3))) \in \{0, 3, 6, 9, 12\} \tag{4}$$

3.1.2 Nidrā Pillar Formulation (Max 33 Points)

$$S_{\text{Nidrā}} = \min(33,\ P_{\text{sleep}} + P_{\text{consistency}} + P_{\text{quality}}) \tag{5}$$

- **Continuous Sleep Duration Curve ($P_{\text{sleep}} \in [0, 15]$):** In accordance with *Gita* 6.17, both sleep deprivation and Tamasic excess are penalized:

$$P_{\text{sleep}}(h) = \begin{cases} \text{round}\left(\frac{h}{7.5} \times 15\right), & 0 \le h \le 7.5 \quad \text{(+1 pt per 0.5h step)} \\ 15, & 7.5 < h \le 8.5 \quad \text{(Peak restorative window)} \\ \max(6, \text{round}(15 - 2(h - 8.5))), & h > 8.5 \quad \text{(Oversleeping decay)} \end{cases} \tag{6}$$

- **Tactical Split-Sentry *Kūrma* Override:** When a defense sentinel operates on rotating shifts and selects `watchRestProtocol === 'split_sentry_kurma'`, if duty restricts sleep to $4.5 \le h < 7.5$, the engine awards full consistency ($P_{\text{consistency}} = 8$) and injects a +3 tactical recovery compensation:

$$P_{\text{sleep}}^{\text{kurma}} = \min(15,\ P_{\text{sleep}}(h) + 3) \tag{7}$$

- **Sleep Quality ($P_{\text{quality}} \in \{1, 5, 10\}$):** Restorative = 10, Fragmented/Disturbed = 5, Depleted/Lethargic = 1.

3.1.3 Brahmacharya Pillar Formulation (Max 33 Points)

$$S_{\text{Brahmacharya}} = \min(33,\ P_{\text{energy}} + P_{\text{focus}} + P_{\text{guna}}) \tag{8}$$

- **Energy / Ojas Reserve Slider ($P_{\text{energy}} \in [0, 12]$):** Maps slider $e \in \{1..5\}$ via $(e-1) \times 3 \in \{0, 3, 6, 9, 12\}$.
- **Normalized Focus / Grounding Blocks ($P_{\text{focus}} \in [0, 13]$):** For Stage 1 civilian deep work blocks $B \in [0, 10]$, $P_{\text{focus}} = \min(13, \text{round}(B \times 2.2))$. For Stage 2 defense grounding wins $W \in [0, 6]$, $P_{\text{focus}} = \min(13, \text{round}(W \times 3.25))$.
- **Guna / Sahyog Cohesion ($P_{\text{guna}} \in \{1, 4, 8\}$):** Sattva / Connected = 8, Rajas / Friction = 4, Tamas / Isolated = 1.

3.2 Engine 2: Impulse Interceptor PRQ Engine

Grounded in *Bhagavad Gita* 2.62–63 and *Yoga Sutra* 2.33, the Pratipaksha Resilience Quotient ($P \in [0, 100]$) measures real-time impulse deceleration. Let $E_{\text{today}} = E_{\text{int}} \cup E_{\text{slip}}$ denote today's recorded urge events, and define the activity indicator $\delta_{\text{int}} = \mathbb{I}(|E_{\text{today}}| > 0)$.

- **Standby Rule:** If $\delta_{\text{int}} = 0$, the user recorded no urges or slips today. $P = 0$, and the module is marked as Standby (excluded from Chariot Composure weighting).
- **Active Slip Penalization:** If $\delta_{\text{int}} = 1$ and $|E_{\text{int}}| = 0$ (the user only logged Honest Slips today), $P = 0$ and **is actively counted**, directly pulling down the composite Chariot Composure.
- **Intensity-Weighted Dynamic Ratio:** When $|E_{\text{int}}| > 0$, each event is weighted by its recorded intensity slider $I \in \{1..5\}$:

$$W_+ = \sum_{i \in E_{\text{int}}} (0.7 + 0.3 I_i) \tag{9}$$

$$W_- = \sum_{j \in E_{\text{slip}}} (0.8 + 0.35 I_j) \tag{10}$$

$$R_{\text{raw}} = \frac{W_+}{W_+ + W_-} \tag{11}$$

$$F_{\text{intensity}} = \min\left(1.0,\ 0.78 + 0.045 \sum_{i \in E_{\text{int}}} I_i\right) \tag{12}$$

$$P = \max(0, \min(100, \text{round}(100 \times R_{\text{raw}} \times F_{\text{intensity}}))) \tag{13}$$

3.3 Engine 3: Bhāva Compass Emotional Conditioning Engine

Grounded in *Patanjali Yoga Sutra* 1.33 (*Chitta Prasādanam*), the Bhāva Engine assesses affective equilibrium ($B \in [0, 100]$). In contrast to older models that defaulted to an arbitrary 75, **BrahmaPath** anchors the neutral baseline at exactly 50:

$$B = \max\left(0,\ \min\left(100,\ 50 + 10N_{\text{pos}} - 12N_{\text{unresolved}} + 4N_{\text{resolved}}\right)\right) \tag{14}$$

where $N_{\text{pos}} \in \{0..3\}$ is the count of positive psychological buffers (capped at +30), $N_{\text{unresolved}}$ is the count of unaddressed interpersonal frictions (penalized at −12 each), and N_{resolved} is the count of successfully resolved negative frictions (rewarded at +4 each, reflecting emotional resolution). The theoretical ceiling of 100 is earned when $N_{\text{pos}} = 3$ and $N_{\text{resolved}} = 5$ ($50 + 30 + 20 = 100$). The activity indicator is $\delta_{\text{bhava}} = \mathbb{I}(N_{\text{pos}} + N_{\text{unresolved}} + N_{\text{resolved}} > 0)$.

3.4 Engine 4: Master Chariot Composure Index and Multi-Gate State Transitions

Drawing upon the chariot allegory of the *Katha Upanishad* (1.3.3–4), where the Self is the Rider, the Intellect (*Buddhi*) is the Charioteer, the Mind (*Manas*) is the Reins, and the Senses (*Indriyas*) are the Horses, the Master Composure Index ($C \in [0, 100]$) dynamically synthesizes physical, impulse, and emotional telemetry:

3.4.1 Dynamic Branch Weighting

$$C = \begin{cases} 0, & \text{if } k = 0 \wedge \delta_{\text{int}} = 0 \wedge \delta_{\text{bhava}} = 0 \quad \text{(Dawn Inactivity)} \\ \text{round}(0.65Y + 0.35B), & \text{if } \delta_{\text{int}} = 0 \quad \text{(Interceptor on Standby)} \\ \text{round}(0.50Y + 0.25P + 0.25B), & \text{if } \delta_{\text{int}} = 1 \quad \text{(Interceptor Active)} \end{cases} \tag{15}$$

where $k \in \{0, 1, 2, 3\}$ denotes the number of distinct Sadhana pillars explicitly logged today.

3.4.2 Earned Multi-Gate State Thresholds

To prevent artificial score inflation (such as logging a single morning pillar turning the entire interface green), the Chariot State transitions through four earned gates:

1. **Resting (Slate Grey Artwork):** Triggered when the user has logged zero activity across all modules today ($k = 0 \wedge \delta_{\text{int}} = 0 \wedge \delta_{\text{bhava}} = 0$), or when severe Tamasic depletion occurs (Guna = Tamas $\wedge\, C < 42$).
2. **Galloping (Red/Terracotta Warning):** Triggered when $C < 50$, OR when $k = 1 \wedge C < 58$. This strict conditional barrier guarantees that logging one incomplete pillar cannot falsely jump the Chariot to Brisk or Calm.
3. **Brisk (Amber/Orange Dynamic Steering):** Triggered when $C \geq 50$ and the state does not meet all three simultaneous criteria for Calm.
4. **Calm (Emerald Green — Sthitaprajña Poise):** Requires satisfying **all three conditions simultaneously**:

$$\text{State} = \text{Calm} \iff (C \geq 75) \ \wedge \ (k \geq 2) \ \wedge \ (N_{\text{unresolved}} = 0) \tag{16}$$

3.5 Engine 5: Non-Punitive Momentum Score and Tamasic Drift Modeling

Commercial applications apply binary streak resets: missing a single day resets a 30-day streak to 0. **BrahmaPath**'s Analytics Vault models habit inertia as physical momentum ($M \in [0, 100]$). For an unlogged day, the engine applies a gentle −5% natural Tamasic settling drift:

$$M_t = \max(0, \ M_{t-1} - 5) \tag{17}$$

Across an active N-day trajectory window ($N \leq 7$), the composite momentum is computed using a recency-weighted linear filter where recent days carry heavier mathematical weight ($w_i = i$):

$$M_{\text{composite}} = \text{round}\left(\frac{\sum_{i=1}^{N} i \cdot Y_i}{\sum_{i=1}^{N} i}\right) \tag{18}$$

4 Behavioral Simulation and Mathematical Verification Data

To verify the mathematical stability and real-time user experience of the upgraded engines, comprehensive end-to-end simulations were executed.

4.1 Chronological Daily Routine Walkthrough

Table 2 documents the step-by-step telemetry of an undergraduate student user across an entire waking day, demonstrating how the composite Composure Index and visual Chariot gait dynamically respond to biological logging, urge slips, cognitive reframing, and evening emotional resolution.

Table 2: Full-Day Chronological Simulation of User Behavioral Telemetry

Time & User Event	Pillars (k)	Yukta (Y)	Bhāva (B)	Interceptor (P)	Composure (C)	Chariot State
07:00 AM — App opened; zero logs	0/3	0	50	Standby ($\delta = 0$)	18	**Resting (Slate)** *Dawn inactivity rule*
07:30 AM — Logs Nidrā (7.5h Restorative = 33 pts)	1/3	33	50	Standby ($\delta = 0$)	39	**Galloping (Red)** *1 pillar cannot turn Brisk*
12:30 PM — Logs Brahmacharya (28 pts) + 1 Pos. Bhāva	2/3	61	60	Standby ($\delta = 0$)	61	**Brisk (Orange)** *Crosses ≥ 50 with 2 pillars*
03:45 PM — Urge Slip (Intensity 4; Honest Slip)	2/3	61	60	0% ($\delta = 1$)	46	**Galloping (Red)** *Slip drops state*
05:00 PM — Reframe (Intercepts urge via Pratipaksha)	2/3	61	60	52% ($\delta = 1$)	59	**Brisk (Orange)** *Recovers from Red*
09:00 PM — Evening Āhāra (29 pts) + 2nd Pos. Bhāva + 2nd Intercept	3/3	90	70	76% ($\delta = 1$)	82	**Calm (Emerald)** *$C \geq 75, k = 3$, 0 frictions*

4.2 Cross-Stage Architectural Schema Comparison

Table 3 details the parameterization differences between Stage 1 (Civilian) and Stage 2 (Defense), demonstrating why the system maintains strict schema isolation.

4.3 Monotonic Sensitivity and Loophole Hunt Verification

Extensive boundary testing confirmed the following properties:

- **Screen Time Extremes:** At $M_{\text{digital}} = 0$, $P_{\text{screen}} = 12$; at $M_{\text{digital}} = 1{,}440$ (24 hours), $P_{\text{screen}} = 0$. Outputs remain strictly bounded and non-NaN.
- **Sleep Hours Curve:** $P_{\text{sleep}}(0) = 0$; $P_{\text{sleep}}(7.5) = 15$; $P_{\text{sleep}}(16) = 6$ (minimum taper floor preserved).
- **Bhāva Scoring Bounds:** Five positive buffer cards cap strictly at 80 (50+30). Six unresolved frictions clamp strictly at 0 (preventing negative integers). Three positives plus five resolved frictions yield exactly 100.

Table 3: Cross-Stage Architectural Parameter and Schema Comparison

Subsystem	Stage 1: Seeker (Civilian Rhythms)	Stage 2: Sentinel (Defense Terrains)
Global Context	Context Key: `lifeRhythm` (Learning & Study, Work & Enterprise, Home & Quiet Living)	Context Key: `deploymentEnvironment` (High-Altitude Cold, Active Border Picket, Cantonment Peace Station)
Sadhana Hub (Trayopastambha)	10 UI Inputs → 10 Schema Fields Nutritional Quality (Sattvic/Rajasic/Tamasic), Bedtime Schedule Consistency (±30m), Study Blocks (0–10 scale).	10 UI Inputs → 13 Schema Fields Adds: `watchStimulantProfile`, `watchRestProtocol` (Split-Sentry Kurma), `sahyogCohesion`, Personal Grounding Wins (0–6 scale).
Impulse Interceptor	5 UI Inputs → 5 Schema Fields Urge Intensity (1–5), 5 Context Triggers, 1-Tap Pratipaksha Reframe Cards, 8 Civilian Micro-Swaps.	7 UI Inputs → 8 Schema Fields Adds: *Vāṅ*-maya Tapas 10-Minute Speech Shield (BG 17.15), 8 Terrain-Safe Somatic Micro-Swaps.
Bhāva Compass	8 Cards (3 Positive Buffers + 5 Negative Frictions) Academic comparison, imposter syndrome, toxic project partner, social isolation, academic overwhelm.	8 Cards + 15 Interactive Checkable Sub-Steps Homefront Reassurance Script (BG 2.14), Swallowed Command Heat (BG 2.56), De-Armoring (BG 12.13), Tele-MANAS (14416).

- **Tridosha Baseline Regularization:** An earlier revision of 'gunaDoshaEngine.ts' normalized acute Dosha points from zero, causing an isolated 4-point disturbance to register as an alarming "100% Vata Imbalance." Regularizing the algorithm with the classical *Sama-Dosha* constitutional baseline ($V_0 = 12, P_0 = 12, K_0 = 12$) restored monotonic decrease in Vata as sensory mindfulness increases ($50\% \to 48\% \to 46\% \to 43\% \to 40\%$).

5 Deep-Dive into Feature Curation Across All Six User Groups

5.1 Foundational Sankalpa Anchor and Vedic Intention Space

Located directly above the Daily Gita Shloka on both Home Dashboards ('ChariotHome.tsx' and 'SentinelDashboard.tsx') is the permanent **Foundational Sankalpa Card**. This card displays the immutable pledge accepted by the user on Day 1, grounded in *Bhagavad Gita* 6.5 ("Let a man uplift himself by his own Self; let him not degrade himself, for the Self alone is the friend of the self"). The card links to the **Vedic Sankalpa Intention Space**, which separates present-tense 24-hour *Immediate Intentions* from multi-week *Horizon Goals*, illuminated by a dedicated Shastra Lens entry based on *Bhagavad Gita* 2.41 (*Vyavasāyātmikā Buddhi*) and *Śukla Yajurveda* 34.6 (*Śiva-Saṅkalpa Sūktam*).

5.2 Sadhana Hub Contextual Customization

The Sadhana Hub dynamically parameterizes its telemetry labels, clinical guidelines, and helper text across all six groups:

- **Learning & Study:** Evaluates 'Dedicated Study Sessions (0–10)', tracks late-night snack and scrolling carryover, and provides visual feedback on focus blocks.
- **Work & Enterprise:** Evaluates 'Deep Work & Key Priorities (0–10)', tracks meeting caffeine dependence, and prompts an evening boundary shutdown.
- **Home & Quiet Living:** Evaluates 'Mindful Daily Anchors (0–10)' (morning walks, caregiving, domestic prayer, reading), warm gentle digestion, and peaceful afternoon rhythm.
- **High-Altitude / Cold:** Evaluates hypoxia appetite suppression (*Agni-Māndya*) with immediate warm-water and soup Battle-Buddy popups, fiberglass-hut broken rest, and cold-weather Ojas conservation.
- **Active Border / Picket:** Evaluates 6-cup sweet watch-tea survival, split-sentry Kurma sensory power-downs, and 5G border phone call digestion.
- **Cantonment / Peace:** Evaluates parade inspection fatigue, barracks hierarchy friction, and regimental buddy cohesion.
- **Self-Anchored Screen Time Helper Box:** Positioned beneath the screen-time slider, an inline info box educates users on their personal baseline snapshot, linking directly to settings for transparent adjustments.

5.3 Impulse Interceptor: 48 Physiologically Grounded Shreyas Micro-Swaps

To eliminate configuration cognitive load, manual micro-swap configuration was removed from settings. Instead, **BrahmaPath** curates **eight rich, physiologically detailed cards per group (48 total)** directly within the Impulse Interceptor interface. Each card incorporates:

1. A bold **Sanskrit + Action Title** (e.g., *Pranayama - Extended Exhale*, *Trataka - Fixed Gazing*, *Kurma Pratyahara - Warm Friction Eye Cupping*, *Vyayama - Boot & Finger Circulation Flex*).
2. A rigorous **two-sentence vagal/neurological explanation** explaining how the action stimulates parasympathetic down-regulation or resets ocular motor strain.
3. An uppercase **Modality Tag** (`INVISIBLE`, `OPTICAL RESET`, `SOMATIC WARMTH`, `KINETIC RELEASE`, `SPACE SHIFT`), allowing the user to select interventions suitable for an open library, a boardroom, or a sentry bunker.

5.4 Bhāva Compass and Interactive Sahyogi Battle-Buddy Kits

The Bhāva Compass implements Patanjali's four-fold affective counter-attitudes (*Maitrī*, *Karuṇā*, *Muditā*, *Upekṣā*). In Stage 2, traditional lectures are replaced with five interactive, three-step checkable Battle-Buddy kits:

1. **Homefront Helplessness & Leave Delay (BG 2.14):** Validates the soldier's love for family, provides an instant ready-to-speak **2-Sentence Homefront Reassurance Script** ("I am on duty guarding our post, but my heart is right there with you. Handle today gently, and I will stand with you the moment my leave opens"), and prompts delegating one practical local task to a relative or buddy.
2. **Swallowed Command Heat (BG 2.56):** Guides the soldier to separate saluting the rank from absorbing a superior's foul temper into their personal bunk.
3. **Uniform Armor De-Armoring (BG 12.13):** A 20-second pre-call breathing ritual to remove defense hypervigilance before speaking to family.
4. **Post-Patrol Alertness (BG 6.26):** A tactile boot-unlacing and weapon-stabling "Gear-Off, Radar-Off" protocol.
5. **Field Numbness (BG 6.5):** A zero-pressure bunker reset prompting one tiny act of warmth without guilt.

All negative friction cards across civilian and defense stages embed the permanent, non-intrusive clinical safeguard: **Tele-MANAS (14416)**, Regimental Medical Officer (RMO), and Mansik Sahayata helpline.

6 Native Android OS Notification Engine and Full-Stack Architecture

6.1 Root-Cause Analysis of Capacitor Repetition Scheduling Failure

During hardware Android deployment, an acute scheduling bug was uncovered in Capacitor's native notification bridge (`LocalNotificationManager.java`). When developers configure a daily recurring alarm by combining `every: 'day'` with `on: { hour, minute }`, the underlying Java implementation prioritizes the interval trigger over the match object:

$$\text{if (trigger.getEvery() != null)} \longrightarrow \text{ignores trigger.getOn()} \tag{19}$$

Consequently, rather than firing at 08:30 and 13:30, the Android OS fired all scheduled daily notifications simultaneously at the exact millisecond of app launch, corrupting the circadian reminder rhythm.

6.2 7-Day Rolling Explicit Timestamp Queue Implementation

BrahmaPath resolved this native limitation by engineering a **7-Day Rolling Explicit Timestamp Queue**. Instead of relying on buggy recurring flags, the background engine calculates concrete calendar timestamps (`at: new Date(year, month, day, hour, min)`) across the next rolling 168 hours. Each notification is registered through Android's `AlarmManager` with `allowWhileIdle = true` on an Importance-5 channel, leveraging the native `USE_EXACT_ALARM` permission to bypass Android OS battery-saver Doze mode.

6.3 Lockscreen Geometry and Notification Anchors

To prevent Android's system UI from truncating notifications on standard mobile lockscreens, all notification copy was strictly engineered under **95 characters (2-line lockscreen ceiling)**. The system schedules seven daily prompts:

- **Four Fixed Anchors:** 08:30 Morning Sankalpa, 13:30 Midday Trayopastambha Pause, 18:30 Evening Sensory Reset, and 21:30 Night Rest Reflection.
- **Three Randomized Mindfulness Windows:** Jittered intervals (10:15–12:15 Deep Work Focus, 15:45–17:30 Emotional Equilibrium Check, and 20:00–21:15 Digital Curfew Warning) that counteract predictable user habituation.

7 Discussion, Future Work, and Why BrahmaPath Deliberately Rejects an AI Chatbot

7.1 The Architectural Rejection of Large Language Model Chatbots

A central design decision of **BrahmaPath** is its explicit rejection of conversational LLM chatbot interfaces (e.g., "Chat with Lord Krishna AI"). While conversational agents represent a prominent commercial trend in digital wellness, our architectural analysis revealed four decisive objections:

1. **The "Why Not Just Use ChatGPT?" Principle:** If a mental health or philosophical application is merely a prompt-engineered wrapper around an external cloud LLM, the product possesses zero technological defensibility and zero user retention value. A user encountering an obstacle will simply query ChatGPT, Claude, or Gemini directly on their device. A true digital therapeutic must provide structured psychometric telemetry, closed-loop state tracking, and somatic interventions that an unanchored text box cannot replicate.
2. **Theological Hallucination and Clinical Safety:** Commercial LLMs are probabilistic token predictors that frequently fabricate Sanskrit verses, conflate conflicting schools of Indian philosophy, or generate unsafe, invalidating advice during acute psychiatric distress. In contrast, every scriptural verse, transliteration, exegesis, and cognitive reframe in **BrahmaPath** is 100% deterministic, human-curated, and reviewed against classical *Sankara Bhashya* commentaries.
3. **100% Offline Military OPSEC Compliance:** Frontline defense deployments occur in high-security, forward operational pickets (Siachen, LOC, LAC) characterized by zero commercial internet access, strict radio silence, and stringent OPSEC protocols. Cloud-based LLM APIs cannot function in disconnected environments and violate military data sovereignty by transmitting personnel emotional telemetry to third-party commercial clouds. **BrahmaPath** operates strictly offline, storing zero data off-device.
4. **Action (*Sadhana*) Over Endless Intellectualizing:** Ancient Indian philosophy emphasizes that mental transformation arises from disciplined, continuous action (*Abhyāsa* and *Kriyā*), not passive intellectual debate. Endless text conversations with a chatbot enable cognitive avoidance, allowing users to talk about their stress rather than putting the phone face down, regulating their breath, and restoring biological equilibrium.

7.2 Future Empirical Validation and Clinical Roadmap

Future work is oriented toward three clinical and technological milestones:

1. **Circadian Schedule Adaptation:** Expanding our six micro-environments with dynamic seasonal and schedule-aware adjustments (e.g., student examination weeks versus semester breaks; military pre-deployment, forward picket, and post-leave transition phases).

2. **Regional Audio-Guided Pratyahara:** Integrating offline, native-language voice tracks (in Hindi, Punjabi, and regional Indian languages) for tactile *Kūrma* eye-cupping and down-regulation breathing, tailored for exhausted sentries and senior citizens.

3. **Formal Longitudinal Clinical Trials:** Conducting randomized controlled trials (RCTs) in collaboration with the **IKSMHA Centre at IIT Mandi**, measuring pre/post intervention shifts in the *Mysore Triguna Scale*, the *General Health Questionnaire (GHQ-12)*, and objective sleep latency across student and paramilitary cohorts.

8 Conclusion

BrahmaPath demonstrates that classical Indian Knowledge Systems contain sophisticated, mathematically formalizable cognitive and physiological models uniquely suited for modern digital therapeutics. By operationalizing the *Katha Upanishad*'s chariot allegory, the *Charaka Samhita*'s tripartite biological balance, the *Bhagavad Gita*'s impulse deceleration cascade, and *Patanjali*'s emotional buffering framework into a deterministic, offline-first state machine, the platform eliminates the punitive failures of Western habit trackers while avoiding the passive textual limitations of existing cultural apps. Through dual-stage schema isolation, six micro-curated vocational cohorts, 48 somatic micro-swaps, and a 7-day rolling native OS alarm queue, **BrahmaPath** delivers a publication-grade, clinically grounded architecture providing calm, focus, and resilience for both civilian life mastery and frontline defense equilibrium.

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